

Optimized Random Forest Model for Remaining Useful Life Prediction of Experimental Bearings

Muhammad Gibran Alfarizi , Bahareh Tajiani, Jørn Vatn, and Shen Yin , *Senior Member, IEEE*

Abstract—Bearings are essential to the reliable operation of rotating machinery in manufacturing processes. There is a rising demand for accurate bearing remaining useful life (RUL) predictions. The data-driven technique for predicting RUL of bearing has demonstrated promising prospects to facilitate intelligent prognostics. This article proposes a new data-driven prediction framework for bearing RUL utilizing an integration of empirical mode decomposition, random forest (RF), and Bayesian optimization. The proposed framework consists of two main phases: 1) feature extraction and 2) RUL prediction. The first phase of this framework focused on decomposing the empirical input signals using empirical mode decomposition into distinct frequency bands to filter out irrelevant frequencies and determine the fault characteristics of the signals. In the second phase, the RUL prediction is then carried out by an RFs-based model with its hyperparameters tuned by Bayesian optimization. The proposed approach is validated using datasets obtained from an actual run-to-failure experiment of roller bearings. The experiment results significantly improved compared to the standard data-driven and stochastic approaches.

Index Terms—Bayesian optimization, empirical mode decomposition (EMD), random forest (RF), remaining useful life (RUL) prediction, roller bearings.

I. INTRODUCTION

ROLLING bearings are vital in ensuring a stable operation in many industrial mechanical drive systems, such as aero engines, high-speed trains, and wind turbines. However, the operation of bearings may encounter many unexpected failures due to lack of lubrication, overload, and inappropriate installation. To avoid undesired consequences due to bearings failure, researchers have developed several remaining useful life (RUL) prediction methods to estimate the failure of the rolling bearings in advance and to schedule proper maintenance actions [1], [2]. RUL prediction is an efficient tool to improve the overall system's reliability and reduce maintenance expenses. However, the challenge remains in describing the degradation trend of rolling bearings due to unknown external circumstances,

such as ambient noise and rotational speed, which can produce a significant discrepancy between actual rolling bearing life and theoretical calculating value.

In recent years, many approaches have been proposed to predict the RUL of bearings. These approaches can be mainly classified into model-based, data-driven, or a combination of both. Model-based approaches use mathematical functions or mappings to represent the physical nature of the failure [3]. Most model-based approaches use the crack growth curves to describe the degradation of rolling bearing, such as the Marco–Starkey theory [4], where statistical estimation methods are applied, e.g., extended Kalman filtering (EKF) [5], to update the prediction over time. Some researchers have favored stochastic processes within model-based approaches because random errors in measurements, working environment uncertainties, and stochastic dynamics in the degradation process could be considered [6]. However, several other variables will influence the final prediction findings, such as fatigue theory, computation method, materials, fracture, and vibration, resulting in a lack of extensibility. Therefore, an increasing number of researchers pay more attention to the data-driven approaches, which use historical failure data to build the degradation model without knowing the physical failure mechanism of the system. A recent review of data-driven methods for RUL prediction can be found in [7].

Data-driven approaches for bearing RUL prediction can be broadly divided into health indicator (HI)-based and direct prediction methods. Using similarity-based interpolation techniques, the first category generates a synthetic HI that depicts the level of system degradation from sensor data. For instance, Huang et al. [8] employed a similarity weighting technique based on Euclidean distance for RUL estimation. They also produced confidence intervals for the predicted RULs using an adaptive kernel density estimation method for managing uncertainty. Likewise, Wang et al. [9] suggested HI curve modeling for online RUL prediction based on sparse Bayesian learning. Yu et al. [10] used a similarity-based HI curve matching method for RUL prediction and a bidirectional recurrent-neural-network-based autoencoder to construct HIs. Although the capability of this group of ways to incorporate new instances is beneficial, the system degradation patterns could be distorted by the data-fusion techniques used to simulate HIs [11].

Machine learning models are explicitly utilized for RUL prediction in direct prediction methods. According to recent studies, HIs are just rescaled RULs and are directly predicted rather than RULs [12], [13]. Recent advances in deep learning have led to the development of numerous direct prediction algorithms.

Manuscript received 22 June 2022; revised 9 August 2022 and 22 August 2022; accepted 4 September 2022. Date of publication 13 September 2022; date of current version 24 May 2023. Paper no. TII-22-2667. (Corresponding author: Muhammad Gibran Alfarizi.)

The authors are with the Department of Mechanical and Industrial Engineering, Faculty of Engineering, Norwegian University of Science and Technology, 7491 Trondheim, Norway (e-mail: muhammad.g.alfarizi@ntnu.no; bahareh.tajiani@ntnu.no; jorn.vatn@ntnu.no; shen.yin@ntnu.no).

Color versions of one or more figures in this article are available at <https://doi.org/10.1109/TII.2022.3206339>.

Digital Object Identifier 10.1109/TII.2022.3206339

Using a deep belief network, Deutsch and He [14] have created RUL prediction models for gears. Li et al. [15] showed that convolutional neural networks perform better than conventional shallow machine learning models when used to predict the RULs of aircraft engines. Wu et al. [16] show that long short-term memory can predict the RULs of aviation engines.

Despite the effectiveness of deep learning models in various prognostics and health management tasks, the lack of datasets makes it difficult to increase the RUL prediction performance [17]. A significant number of parameters must be modified during the training phases of deep learning models. Deep learning models can offer excellent prediction performance on test samples with good generalization abilities when sufficient data are available. In the case of inadequately labeled historical data, training may result in unsatisfactory model parameters that only apply to the information upon which models were trained [18]. In addition, deep learning requires relatively high computational cost and lengthy training time [19]. Other algorithms, such as the random forest (RF) method, can be utilized to overcome the drawbacks of deep learning.

Developed by Breiman (in 2001) [20], RF is an ensemble machine-learning technique. It is a bagging-based approach that integrates many predictors (regression trees) generated randomly using the bootstrapping method. Multiple predictors can minimize complexity and yield superior performance compared to a single predictor. RFs can handle discrete and continuous variables without making assumptions about their distribution and are not susceptible to overfitting [21]. Moreover, the RFs approach minimizes complexity in the following three different ways.

- 1) It keeps complexity low by constructing submodels from sample subsets using the bootstrap.
- 2) It keeps computation speedy by using a parallel ensemble to create submodels on each sample.
- 3) It has fewer hyperparameters to tune in comparison to deep learning [22].

The hyperparameters must be determined optimally to predict the RUL of bearings using RFs. Hyperparameter optimization identifies a set of hyperparameters that minimizes a predetermined loss function when applied to a group of independent data. One approach to optimizing hyperparameters is the grid search approach, which is an exhaustive search over a manually chosen subset of a learning algorithm's hyperparameters space. Another approach, random search, is an improvement relative to grid search because it replaces the exhaustive enumeration of all combinations by randomly selecting them. When only a small number of hyperparameters influences the machine learning algorithm's performance, it can outperform grid search [23]. However, these two approaches are relatively inefficient since they do not select the next hyperparameters to assess depending on the outcomes of the prior evaluations. Previous evaluations do not influence the grid and random searches. They frequently spend a lot of time analyzing "bad" hyperparameters.

For noisy black-box functions, there is a global optimization technique called Bayesian optimization [24]. The Bayesian approach remembers the outcomes of earlier analyses, unlike random or grid search. Using these data, they create a probabilistic model that connects hyperparameters to the probability

of a score on the objective function $P(\text{score} \mid \text{hyperparameters})$. Bayesian hyperparameters optimization is used because it can identify better model configurations in fewer iterations than random search by assessing hyperparameters that appear more promising based on the previous iteration.

The main objective of this article is to assess how accurately the proposed approach can predict the RUL of rolling bearings. The Bayesian optimization algorithm is used to optimize the hyperparameters of the RFs model. To validate the performance of the proposed method, the authors conducted a run-to-failure experiment of rolling bearings and collected the raw acceleration signals. The natural signals are then decomposed into several HIs using the empirical mode decomposition (EMD) [25]. The most significant advantage of the EMD method is that it is adaptive and data-driven, without the need for *a priori* basis function selection for signal decomposition. In addition, EMD, as a widely used signal processing approach, can also analyze nonlinear and non-stationary vibration signals. These HIs are then selected using feature importance from the RFs algorithm and inputted to the RFs-based Bayesian model. Its performance is compared with support vector regression (SVR), LASSO regression, artificial neural networks (ANNs), and conventional RFs. Additionally, the performance of the proposed model is compared with a model-based approach using the Wiener process.

The main contributions of this article can be summarized as follows.

- 1) A framework to construct HIs is proposed by applying EMD and selecting them using feature importance to determine the best feature for RUL prediction.
- 2) A novel data-driven approach is proposed by utilizing the proposed HI framework, RF, and Bayesian optimization to offer precise RUL prediction to overcome the drawbacks of deep learning (e.g., high computational cost and lengthy training time) and improve prediction accuracy.
- 3) The proposed RUL prediction approach is verified by real-world datasets obtained from a run-to-failure experiment of rolling bearings. Furthermore, the performance of the proposed model has been investigated and compared with other data-driven and model-based approaches for RUL prediction of bearings.

The remainder of this article is organized as follows. Section II presents the proposed framework and the preliminaries of our approach. Section III describes the data used in the case study. Section IV discusses the performance of the proposed model in real-world datasets. Section V concludes this article.

II. PROPOSED METHOD

A. Framework of the Proposed Method

Fig. 1 depicts the framework of the proposed method. The initial phase of this framework is focused on feature extraction, in which each of the gathered raw acceleration signals is decomposed into distinct frequency bands to filter out irrelevant frequencies and determine the fault characteristics of the signals. The signal decomposition is performed with EMD. The EMD decomposition is based on the variation in frequency over the bearings' lifespan. The statistical features as indicators (i.e.,

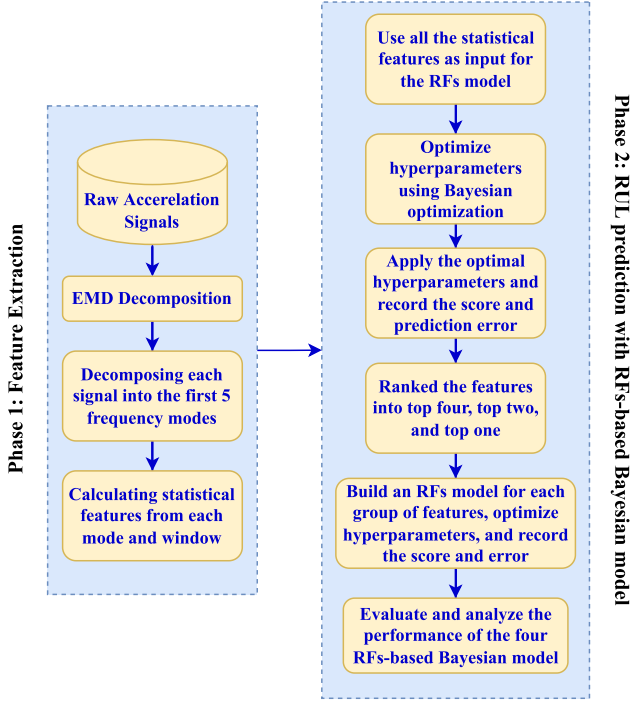


Fig. 1. Framework of the proposed method.

kurtosis, RMS, crest factor, shape factor, impulse factor, skewness, and mean value) are calculated for each intrinsic mode function (IMF) to compare the frequency bands and determine which feature from which band is suitable for predicting RUL of bearings. The EMD decomposition is further elaborated in Section II-B.

The second phase is related to the RUL prediction. The bearing datasets have been obtained by an experimental laboratory with the same operating condition. The statistical features from other bearings' datasets are used as input basis for the RFs-based model to predict the RUL of the current bearing of interest. The hyperparameters of concern (the number of trees in the forest, the maximum depth of the tree, and the minimum number of samples required to be at a leaf node) of every RFs-based model are optimized using Bayesian optimization. The features are then ranked and divided into four groups based on their importance, which results in groups containing all, top four, top two, and top one features. The RFs-based model considers every feature group as the input to predict the RUL of the current bearing of interest. The RUL prediction of every bearing is evaluated based on its score and prediction error. The details of each step will be further elaborated in Section II-F.

B. Empirical Mode Decomposition

Traditional signal-processing techniques assume that the signals are linear or stationary. Thus, they may result in false information and erroneous results. However, among various time-frequency methods, the Hilbert–Huang transform (HHT) is an empirical approach that can analyze stationary and nonstationary signals for fault detection, condition monitoring, and failure prognosis of bearings [25]. HHT is an adaptive data-driven

Algorithm 1: Empirical Mode Decomposition.

Input: Acceleration signal $y(t)$

Identify all local maxima and minima of $y(t)$

for $i = 1$ **to** n **do**

while $j \leq k$ **do**

The upper and lower envelopes, y_u^t and y_l^t , are formed by connecting all local maxima and minima with a cubic spline line, respectively

Calculate the mean of y_u^t , y_l^t , as $m_{i1}^t = (y_u^t + y_l^t)/2$, and $h_{i1}^t = y(t) - m_{i1}^t$

The above sifting process is repeated several times as $h_{ij}^t = h_{i(j-1)}^t - m_{ij}^t$

end while

h_{1k}^t satisfies the definition of IMF and the stopping

condition, SD, where $SD = \sum_{t=0}^T \left(\frac{|h_{1(k-1)}^t - h_{1k}^t|}{h_{1(k-1)}^t} \right)^2$,

it becomes IMF, that is, $c_1 = h_{1k}^t$

A new signal, x_1 , is generated by $x_1 = y(t) - c_1$

Repeat above steps as $x_i = x_{i-1} - c_i$

end for

Output: The original signal is decomposed into n IMFs, and one residual x_n

approach that can also handle the nonlinearity of the signals [26]. The first step in HHT is EMD, which decomposes the signals into different segments called IMFs based on their time-domain characteristics. The process of carrying out an EMD approach on an acceleration signal is summarized in Algorithm 1.

C. Random Forests

The RFs algorithm is based on a collection of predictors influenced by each forest tree's random value. To observe the modeling of RFs, input datasets are extracted from a group and then randomized. The higher success of the RFs algorithm is attributable to its efficient handling of big datasets, rapid operational speed, and lack of predictor overfitting [27].

The forest in RFs is built using the p dimension input vector of $X = x_1, x_2, \dots, x_p$. Then, a set of K trees $\{T_1(x), T_2(x), \dots, T_K(x)\}$ is developed inside the forest. Each tree determines its output value, represented as $\hat{Y}_1 = T_1(X), \dots, \hat{Y}_m = T_m(X)$, where $m = 1, \dots, K$. The overall output of the RFs is calculated by the estimation of an average of all tree predictors, mathematically expressed as

$$\text{Predict}_{\text{RFs}}(X) = \frac{1}{K} \sum_{m=1}^K \hat{Y}_m(X). \quad (1)$$

The training dataset $D = D_1, D_2, \dots, D_n = \{(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)\}$ is taken independently from input and output, where $x_i, i = 1, \dots, n$ denotes the input vector training dataset and $y_i, i = 1, \dots, n$ denotes the output vector training dataset. From the original sample set D , the RF randomly creates K tree sample sets using the bootstrap resampling technique. Approximately one-third of the data in the original sample set D is not drawn in bootstrap samples;

Algorithm 2: Random Forest.

Input: Training samples $D = \{(x_1, y_1), \dots, (x_n, y_n)\}$, testing samples x_t

for $m = 1$ **to** K **do**

Construct the bootstrap sample D_m randomly from the original training set D with replacement

Create a nonpruning decision tree $\hat{Y}_m$ through D_m

Pick n_{try} features at random from N features

Choose the best feature from each node's n_{try} features to split

Split the tree until it reaches its maximum size

end for

Output: A set of decision trees $\{\hat{Y}_m, m = 1, 2, \dots, K\}$.

The decision tree $\hat{Y}_m$ generates the predictor $\hat{Y}_m(x_t)$ for the testing samples. The overall output of the RFs is calculated as in (1)

these data are known as out-of-bag (OOB) data, whereas the remaining data are known as in-bag data. Pseudocode of the RF algorithm [20] is described in Algorithm 2.

As stated previously, the growth of each tree depends on a bootstrap sample containing almost two-thirds of the training data. Testing is done with the remaining training data (OOB). As shown in the following equation, the MSE of the OOB error is derived by deducting the predicted values from the reference values

$$\text{MSE} \approx \text{MSE}^{\text{OOB}} = \frac{1}{N} \sum_{i=1}^N (\hat{Y}(X_i) - Y_i)^2 \quad (2)$$

where $\hat{Y}(X_i)$ denotes the predicted output, Y_i denotes the observed output, and N represents the total number of samples.

D. Bayesian Optimization

Bayesian optimization produces a probabilistic model of the function mapping from hyperparameter values to the objective as evaluated by a validation set when used for hyperparameter optimization. Bayesian optimization is an iterative procedure that involves two fundamental components: 1) a probabilistic surrogate model and 2) an acquisition function to determine the next point to examine. In each iteration, the surrogate model is fitted to all previous observations of the target function. After that, the acquisition function, which utilizes the probabilistic model's predictive distribution, decides the utility of various prospective spots while balancing exploitation and exploration. The acquisition function can be thoroughly tuned because it is inexpensive to compute compared to assessing the costly black-box process.

A visual representation of Bayesian optimization is shown in Fig. 2. After two evaluations, the black line represents the surrogate model's initial estimate, with the related uncertainty shown in gray. The surrogate model falls short of the actual objective function shown in red.

The surrogate function after eight evaluations is shown in Fig. 3. The real objection function is now matched with the surrogate function. In light of this, the hyperparameters that

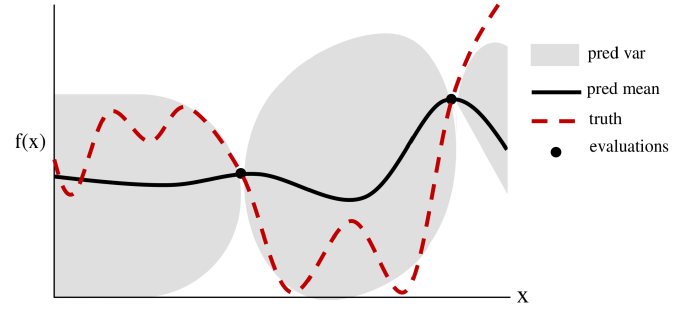


Fig. 2. Bayesian optimization run after two evaluations. Modified from Snoek et al. [28].

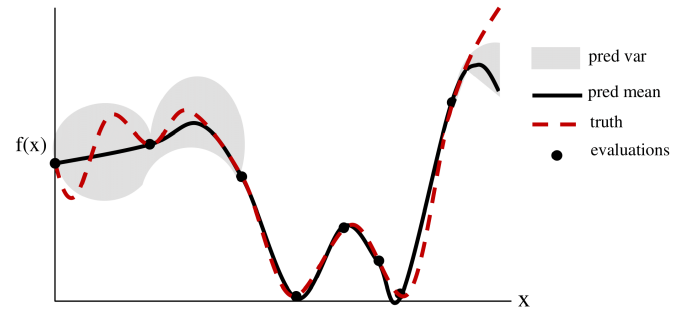


Fig. 3. Bayesian optimization run after eight evaluations. Modified from Snoek et al. [28].

maximize the surrogate will also optimize the actual objective function.

One of the acquisition functions is expected improvement (EI)

$$\text{EI}_{y^*}(x) = \int_{-\infty}^{y^*} (y^* - y)p(y|x)dy \quad (3)$$

where y^* is the objective function's threshold value, the objective function value y is calculated using the hyperparameters x and the surrogate probability model $p(y|x)$ expresses the probability that y will occur given x . The goal is to maximize the EI concerning x .

The tree-structured Parzen estimator applies the Bayes rule, which is expressed as

$$p(y|x) = \frac{p(x|y) * p(y)}{p(x)} \quad (4)$$

to construct a model where the probability of the hyperparameters given the score on the objective function is denoted by $p(x|y)$, which may be written as

$$p(x|y) = \begin{cases} l(x) & \text{if } y < y^* \\ g(x) & \text{if } y \geq y^* \end{cases} \quad (5)$$

where $y < y^*$ denotes an objective function value below the threshold. According to this equation, there are two possible distributions for the hyperparameters: One in which the objective function value is lower than the threshold ($l(x)$) and another in which it is greater ($g(x)$).

Since the $l(x)$ distribution only includes x that resulted in scores below the threshold, draw values of x from this distribution are superior. Bayes' rule also supports it since the EI

Algorithm 3: Bayesian Hyperparameter Optimization.

Input: (hyperparameter space Θ , Target score function $H(\theta)$, max number of evaluation $n_{\max}$)
 Select an initial configuration $\theta_0 \in \Theta$
 Evaluate the initial score $y_0 = H(\theta_0)$
 Set $\theta^* = \theta_0$, $y^* = H(\theta_0)$, and $S_0 = \{\theta_0, y_0\}$
for $n = 1$ **to** $n_{\max}$ **do**
 Select a new hyperparameter configuration $\theta_n \in \Theta$ by maximizing EI in (6)
 Evaluate H in θ_n to obtain a new numeric score $y_n = H(\theta_n)$
 Augment the data $S_n = S_{n-1} \cup \{\theta_n, y_n\}$
 Update the surrogate model
 if $y_n < F^*$ **then**
 $\theta^* = \theta_n$ and $y^* = y_n$
 end if
end for
Output: θ^* and y^*

equation becomes

$$\text{EI}_{y^*}(x) = \frac{\gamma y^* l(x) - l(x) \int_{-\infty}^{y^*} p(y) dy}{\gamma l(x) + (1 - \gamma) g(x)} \propto \left(\gamma + \frac{g(x)}{l(x)} (1 - \gamma) \right)^{-1} \quad (6)$$

which states that the EI is inversely proportional to $\frac{l(x)}{g(x)}$. Therefore, this ratio will be maximized when values for x are taken from $l(x)$, increasing the EI. The variable γ represents the quantile of the negative accuracy scores (observed thus far), used as a cutoff point, and defines the threshold y^* . This user-defined cutoff point is often set at 15% by default.

Before saving the results and scores, the algorithm compares each new set of hyperparameters it proposes with the real objective function. The algorithm builds a probabilistic model that gets better with each iteration by computing $l(x)$ and $g(x)$ using the account of the objective function.

In conclusion, a preliminary estimate of the surrogate function is developed and modified as new information becomes available. When there are enough evaluations of surrogate functions, the objective function will eventually be adequately reflected. The objective function would also be maximized by the same hyperparameters that maximize the EI. The pseudocode for Bayesian hyperparameter optimization is presented in Algorithm 3.

E. Model Evaluation

A score function is defined to evaluate the model's performance by comparing the actual and predicted RUL of bearings. The score function is based on the percent errors of predictions, which is defined as

$$\% \text{Error} = 100 \times \frac{\text{Act RUL} - \widehat{\text{RUL}}}{\text{Act RUL}}. \quad (7)$$

The early prediction of RUL (i.e., cases where $\% \text{Error} > 0$) and late prediction of RUL (i.e., cases where $\% \text{Error} \leq 0$) will be

TABLE I
INITIAL HYPERPARAMETERS AND THEIR SEARCH RANGE

Hyperparameter	Initial Value	Search Space
The number of trees	100	[10:500]
The maximum depth of tree	50	[2:100]
The minimum leaf per tree	1	[1:50]

treated differently. Thus, a penalty function, expressed in (8), is introduced, which penalizes late RUL predictions more than early ones. Late RUL predictions might result in more severe consequences and are more crucial than early ones [29]

$$A_i = \begin{cases} \exp(-\ln(0.5) \cdot (\% \text{Error}/5)) & \text{if } \% \text{Error} \leq 0 \\ \exp(+\ln(0.5) \cdot (\% \text{Error}/20)) & \text{if } \% \text{Error} > 0. \end{cases} \quad (8)$$

Moreover, accurate RUL predictions at the later stage of a bearing's lifetime are more vital than predictions at the early stage. So, higher weights are given to the predictions near the bearing's time to failure. The weight is assigned linearly as the first prediction data point weights 1, and the n th prediction weights n . The score function is then defined as

$$\text{Score} = \frac{\sum_{i=1}^n w_i \times A_i}{\sum_{i=1}^n w_i} \quad (9)$$

where n is the number of samples, and w_i is the weight of i th prediction.

In addition to the score function, the root-mean-square relative error (RMSRE) is also calculated. RMSRE is expressed as

$$\text{RMSRE} = \sqrt{\frac{1}{n} \sum_{i=1}^n \left(\frac{\text{Act RUL} - \widehat{\text{RUL}}}{\text{Act RUL}} \right)^2}. \quad (10)$$

A higher score and lower RMSRE values indicate that the model's predictive abilities for RUL prediction are superior.

F. Framework of the Proposed RFs-Based Bayesian Model

The proposed model's objective is to accurately predict the RUL of rolling bearings. The RFs regression model is responsible for predicting the RUL while Bayesian optimization will optimize the RFs model's hyperparameters to increase its prediction accuracy. The RUL prediction of bearings begins with specifying the input sample, the features, and the hyperparameters' initial values. The initial hyperparameters and their search range are summarized in Table I.

The optimal hyperparameter settings will decrease prediction error [30]. According to the literature, no analytical formula exists to find the optimum set of hyperparameters [31]. Thus, Bayesian optimization is combined with the RFs model to determine the optimal hyperparameter settings, resulting in lower prediction error. The objective of the optimization is to minimize the MSE of the RFs model. The flowchart of the proposed RFs-based Bayesian model is shown in phase 2 of Fig. 1.

The steps involved in phase 2 of the flowchart are explained as follows.

TABLE II
BEARINGS AND THEIR NUMBER OF SAMPLES

Bearing	B1	B2	B3	B4	B5
Number of samples	110	35	146	106	77
Bearing	B6	B7	B8	B9	B10
Number of samples	248	150	143	114	115

- 1) The samples and features are obtained from the feature extraction phase with EMD decomposition using other bearings' data for each bearing. The features used for input are kurtosis, RMS, crest factor, shape factor, impulse factor, skewness, and mean value. The output is the RUL of bearings for every timestep.
- 2) The hyperparameters (see Table I) are set to the initial values and then optimized using Bayesian optimization. It was observed that most of the time, the value of the surrogate function does not improve after 40 evaluations; the maximum number of evaluations is set to 50.
- 3) The optimal set of hyperparameters is applied to the RFs model, and the score and the RMSRE are recorded for model evaluation.
- 4) The features are ranked based on their importance in predicting the RUL of bearings using RFs feature importance and SHAP importance analysis. The best 50% of the features are selected and then ranked again. The process is repeated, and as a result, the features are divided into the top four, the top two, and the top one.
- 5) Build an RFs model for each group of features and repeat steps 2 and 3. In total, there are four different RFs models for each bearing.
- 6) Evaluate and analyze the performance of the four RFs-based Bayesian models.

III. CASE STUDY

A. Data Description

A set of run-to-failure experimental tests on roller bearings is designed to illustrate the effectiveness of the proposed approach. The bearings are operated under similar laboratory conditions with the same motor speed of 2975 r/min. The original data are a batch of acceleration signals in the horizontal direction collected from the bearings' healthy state until the acceleration amplitude crosses the level of 10g where the experiments ceased. The bearings are degraded by contamination, i.e., pouring a mixture of Silicon carbide solid particles and lubricant onto the bearing at regular intervals until the stopping threshold (i.e., 10g) is reached. Table II lists the number of samples of different bearings from a healthy state to a failed state. The details of the data and the vibration setup used to conduct the experiments can be found in [32].

B. Health Indicator

In this section, EMD is implemented to decompose each horizontal acceleration signal over different frequency ranges according to the signal's time-scale characteristics resulting in

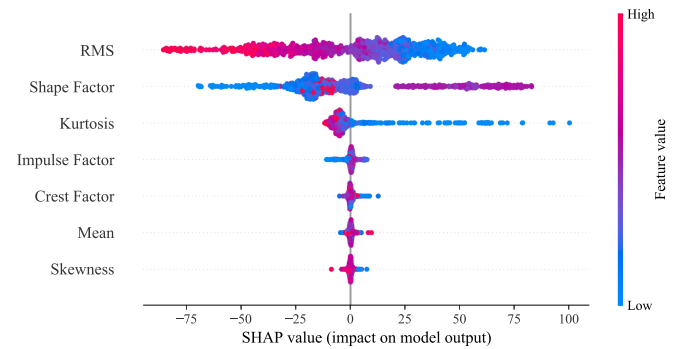


Fig. 4. SHAP variable importance plot for B10.

a series of IMFs. Comparing different IMFs with their feature scales for each signal gives insights into the relevant or informative frequencies and features concerning the bearings' faults. The statistical features will then be calculated from each of the first 5 IMFs of the acceleration signals. The first IMF in EMD is the dominant feature among all with the highest value of fitness function (i.e., metric that combines monotonicity, trendability, and prognosability). It is thus chosen for the HIs. Table III shows how to calculate the statistical features where M is the number of measurements in one sample, c_i is the processed signal measurements obtained by the EMD algorithm, and the mean and standard deviation are m and σ , respectively.

In every bearing, the RMS feature has the highest value in monotonicity, prognosability, and trendability, which means it is the essential feature for RUL prediction. The RFs feature importance and SHAP importance analysis also corroborate this finding, as RMS is the critical feature for every bearing. RMS is also the only feature that visually trends upward as time passes, indicating higher degradation over time until bearing failure. However, the value of RMS in one bearing cannot be compared to other bearings, as the RMS value at failure time is different in every bearing. Thus, the RMS feature must be considered individually.

IV. EXPERIMENT RESULTS AND ANALYSIS

A. Model Results

Fig. 4 shows the SHAP variable importance plot for B10. The plot shows that the RUL prediction depends on the RMS feature. Moreover, the RMS is negatively correlated with the RUL value, which means that the higher the RMS value, the lower the RUL value. This correlation makes sense since higher RMS indicates higher degradation, which means lower the RUL of the bearing. The SHAP variable importance plot shows similar results for other bearings.

Tables IV and V show the score and RMSRE of the RFs-based Bayesian model for every bearing after hyperparameters optimization is applied to every group of features. Fig. 5 shows the RUL prediction for every bearing with the best score.

Tables IV and V show that the proposed model, on average, produces the highest score and lowest RMSRE when using only the top two features. However, the differences in score and error are only significant when using only one feature and insignificant

TABLE III
FORMULA FOR STATISTICAL FEATURES

Feature	Kurtosis	RMS	Crest factor	Skewness
Formula	$K = \frac{\sum_{i=1}^M (c_i - m)^4}{(M-1)\sigma^4}$	$RMS = \sqrt{\frac{1}{M} \sum_{i=1}^M c_i^2}$	$CF = \frac{\max(c_i)}{\sqrt{\frac{1}{M} \sum_{i=1}^M c_i^2}}$	$S = \frac{\sum_{i=1}^M (c_i - m)^3}{(M-1)\sigma^3}$
Feature	Mean	Shape factor	Impulse factor	
Formula	$Mean = \frac{1}{M} \sum_{i=1}^M c_i$	$SF = \frac{\sqrt{\frac{1}{M} \sum_{i=1}^M c_i^2}}{\frac{1}{M} \sum_{i=1}^M c_i }$	$IF = \frac{\max c_i }{\frac{1}{M} \sum_{i=1}^M c_i }$	

TABLE IV
SCORE OF RUL PREDICTIONS FOR EVERY BEARING

Score	B1	B2	B3	B4	B5	B6	B7	B8	B9	B10	Average
All features	0.7621	0.7950	0.9277	0.9533	0.8760	0.9648	0.8809	0.9516	0.9094	0.9461	0.8967
4 features	0.7636	0.7853	0.9307	0.9580	0.8643	0.9634	0.8972	0.9537	0.9164	0.9487	0.8981
2 features	0.7761	0.8597	0.9362	0.9612	0.9329	0.9641	0.8447	0.9603	0.9135	0.9429	0.9092
1 feature	0.9536	0.6414	0.8151	0.8413	0.8096	0.9645	0.8532	0.8266	0.7917	0.8306	0.8328

TABLE V
RMSRE OF RUL PREDICTIONS FOR EVERY BEARING

RMSRE	B1	B2	B3	B4	B5	B6	B7	B8	B9	B10	Average
All features	2.6739	2.2487	1.3654	0.7220	1.1861	0.8493	1.4983	0.8499	1.4479	0.9599	1.3801
4 features	2.6187	2.2637	1.0971	0.6671	1.4759	0.9307	1.3785	0.8547	1.4315	0.8149	1.3533
2 features	2.3878	1.3728	1.2691	0.6417	0.7147	0.8891	1.8001	0.7249	1.2908	1.1332	1.2224
1 feature	0.5278	4.2387	4.3198	3.4316	1.7146	0.8916	1.8849	5.5522	7.3177	2.9226	3.2802

otherwise. The top feature here is the RMS feature. This difference suggests that using multiple HIs is beneficial for predicting RUL values. Many HIs can provide better information about the bearing's health condition than only one. The proposed model also runs efficiently due to the use of Bayesian optimization to update parameters, which avoids the high computational cost and lengthy training time.

There are differences between the proposed method as a data-driven approach and the Wiener process as a stochastic approach. In the Wiener process, the underlying randomness or the probabilistic behavior in the bearings' degradation is considered while the uncertainty of the failure threshold is also considered at each timestep. On the other hand, the proposed method can provide better RUL prediction results regarding score and error without determining a failure threshold.

Fig. 5 shows that the prediction results of the proposed model are close to the actual RUL values in most of the bearings, especially near the end of the bearings' lifetime, where the RUL prediction matters most. Generally, the proposed model performs better when the bearing has an average number of samples (i.e., 100–110 samples) like bearing 1, 4, 9, and 10. However, the model does not give an accurate RUL prediction when the bearing only has a small number of samples (i.e., short lifetime), such as bearing 2 and 5. According to the manufacturer, the average lifespan of such bearings is between 8 and 12 hours. Consequently, bearing 2 and 5 are atypical, and there may be technical problems in conducting experiments. The technical issue may be caused by the accelerometers not being correctly mounted to the bearings, resulting in erroneous signals. In addition, the predictions of bearing 6 perform poorly compared to the other bearings. Its inferior prediction could be due to its higher number of samples (248) than others.

Because the RFs model learned from other bearings datasets to make predictions, it has not seen samples higher than 200 and, therefore, makes outrageous predictions.

Generally, at the beginning of a bearing's lifespan, the model has a more considerable prediction error, and this is due to some uncertainty in the input data. At the middle and deterioration stages, when the bearings are approaching failure, the uncertainty has decreased because more degradation data have been seen and utilized to capture the degradation trend, resulting in a more accurate RUL prediction.

B. Results Validation

A comparison with other data-driven algorithms is made to corroborate the proposed model's performance, including conventional RFs, ANNs [33], SVR [34], and LASSO regression [35]. The hyperparameters of ANN (number of layers, hidden unit, and learning rate), SVR (gamma and C), and LASSO (alpha) are optimized with Bayesian hyperparameters optimization. Moreover, the performance of the proposed model is also compared with a model-based approach using the Wiener process. The average score and RMSRE are recorded for every method and summarized in Table VI. The proposed method results are bolded in Table VI.

Based on the score and RMSRE in Table VI, the proposed method outperforms all other ways to predict the RUL of bearings. Compared to conventional RFs, the proposed model has improved the average score by 3% and reduced the average error by 9%. Furthermore, compared to the Wiener process, the proposed model has enhanced the average score by 6% and reduced the average error by 54%. A higher score and lower

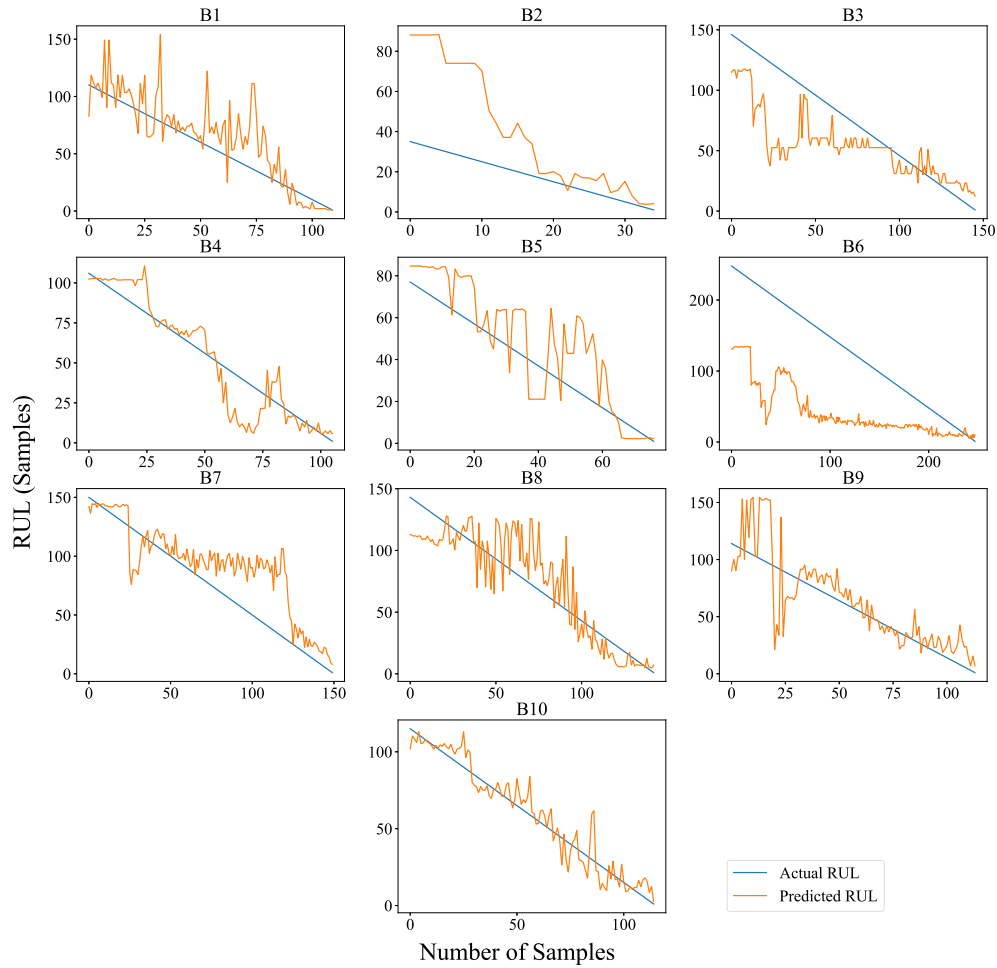


Fig. 5. Best RUL prediction of every bearing.

TABLE VI
SCORE AND RMSRE OF ALL METHODS

Method	Avg. Score	Avg. RMSRE
Proposed method	0.9092	1.2224
Conventional RFs	0.8856	1.3375
ANN	0.7735	6.5496
SVR	0.7581	7.5216
LASSO	0.8196	3.6325
Wiener process	0.8606	2.6301

RMSRE values imply that the model's predictive capabilities for RUL prediction are superior.

V. CONCLUSION

This article proposes a data-driven framework for predicting the RUL of bearings. The framework consists of the feature extraction phase and RUL prediction phase. EMD performed the feature extraction to decompose input signals into different frequency bands to identify signal fault characteristics. Then, an RFs-based model combined with Bayesian hyperparameters optimization was developed to predict the RUL of bearings. A run-to-failure experiment of roller bearings was performed to validate the proposed method. Score function and RMSRE were used to evaluate the method's performance. The proposed

method outperforms other data-driven and model-based approaches in terms of score and error. Additionally, the proposed approach requires relatively low computational cost and fast training time compared to the deep learning approach. However, like any RFs-based method, one has to find a tradeoff between the training time and prediction accuracy. RF is also a black box algorithm, so the results are not easily interpretable, i.e., it does not provide complete visibility into the coefficients like linear regression.

For further research, a more detailed investigation is required to compare the proposed method and the stochastic Wiener process depending on the decision context, the length of the dataset or the bearing's lifetime, accuracy of the prediction results on the two distinct healthy stages, and deterioration stage of bearings, as well as their mathematical complexity and computational time.

REFERENCES

- [1] W. Mao, J. Chen, J. Liu, and X. Liang, "Self-supervised deep domain-adversarial regression adaptation for online remaining useful life prediction of rolling bearing under unknown working condition," *IEEE Trans. Ind. Informat.*, 2022, doi: [10.1109/TII.2022.3172704](https://doi.org/10.1109/TII.2022.3172704).
- [2] Y. Qin, D. Chen, S. Xiang, and C. Zhu, "Gated dual attention unit neural networks for remaining useful life prediction of rolling bearings," *IEEE Trans. Ind. Informat.*, vol. 17, no. 9, pp. 6438–6447, Sep. 2021.

- [3] M. G. Alfarizi, J. Vatn, and S. Yin, "An extreme gradient boosting aided fault diagnosis approach: A case study of fuse test bench," *IEEE Trans. Artif. Intell.*, 2022, doi: [10.1109/TAI.2022.3165137](https://doi.org/10.1109/TAI.2022.3165137).
- [4] L. Cortese, F. Nalli, and M. Rossi, "A nonlinear model for ductile damage accumulation under multiaxial non-proportional loading conditions," *Int. J. Plast.*, vol. 85, pp. 77–92, 2016.
- [5] L. Cui, X. Wang, Y. Xu, H. Jiang, and J. Zhou, "A novel switching unscented Kalman filter method for remaining useful life prediction of rolling bearing," *Measurement*, vol. 135, pp. 678–684, 2019.
- [6] Z. Zhang, X. Si, C. Hu, and Y. Lei, "Degradation data analysis and remaining useful life estimation: A review on wiener-process-based methods," *Eur. J. Oper. Res.*, vol. 271, no. 3, pp. 775–796, 2018.
- [7] Y. Lei, N. Li, L. Guo, N. Li, T. Yan, and J. Lin, "Machinery health prognostics: A systematic review from data acquisition to RUL prediction," *Mech. Syst. Signal Process.*, vol. 104, pp. 799–834, 2018.
- [8] C.-G. Huang, H.-Z. Huang, W. Peng, and T. Huang, "Improved trajectory similarity-based approach for turbofan engine prognostics," *J. Mech. Sci. Technol.*, vol. 33, no. 10, pp. 4877–4890, 2019.
- [9] P. Wang, B. D. Youn, and C. Hu, "A generic probabilistic framework for structural health prognostics and uncertainty management," *Mech. Syst. Signal Process.*, vol. 28, pp. 622–637, 2012.
- [10] W. Yu, I. Y. Kim, and C. Mechefske, "An improved similarity-based prognostic algorithm for RUL estimation using an RNN autoencoder scheme," *Rel. Eng. Syst. Saf.*, vol. 199, 2020, Art. no. 106926.
- [11] R. Khelif, S. Malinowski, B. Chebel-Morello, and N. Zerhouni, "RUL prediction based on a new similarity-instance based approach," in *Proc. IEEE 23rd Int. Symp. Ind. Electron.*, 2014, pp. 2463–2468.
- [12] J. Zhang, Y. Jiang, S. Wu, X. Li, H. Luo, and S. Yin, "Prediction of remaining useful life based on bidirectional gated recurrent unit with temporal self-attention mechanism," *Rel. Eng. Syst. Saf.*, vol. 221, 2022, Art. no. 108297.
- [13] J. Zhang, Y. Jiang, X. Li, M. Huo, H. Luo, and S. Yin, "An adaptive remaining useful life prediction approach for single battery with unlabeled small sample data and parameter uncertainty," *Rel. Eng. Syst. Saf.*, vol. 222, 2022, Art. no. 108357.
- [14] J. Deutsch and D. He, "Using deep learning-based approach to predict remaining useful life of rotating components," *IEEE Trans. Syst., Man, Cybern. Syst.*, vol. 48, no. 1, pp. 11–20, Jan. 2018.
- [15] X. Li, Q. Ding, and J.-Q. Sun, "Remaining useful life estimation in prognostics using deep convolution neural networks," *Rel. Eng. Syst. Saf.*, vol. 172, pp. 1–11, 2018.
- [16] Y. Wu, M. Yuan, S. Dong, L. Lin, and Y. Liu, "Remaining useful life estimation of engineered systems using vanilla LSTM neural networks," *Neurocomputing*, vol. 275, pp. 167–179, 2018.
- [17] C. Sun, M. Ma, Z. Zhao, S. Tian, R. Yan, and X. Chen, "Deep transfer learning based on sparse autoencoder for remaining useful life prediction of tool in manufacturing," *IEEE Trans. Ind. Informat.*, vol. 15, no. 4, pp. 2416–2425, Apr. 2019.
- [18] D. Yoo, H. Fan, V. Boddeti, and K. Kitani, "Efficient k-shot learning with regularized deep networks," in *Proc. AAAI Conf. Artif. Intell.*, vol. 32, no. 1, 2018, pp. 4382–4389.
- [19] C. Ferreira and G. Gonçalves, "Remaining useful life prediction and challenges: A literature review on the use of machine learning methods," *J. Manuf. Syst.*, vol. 63, pp. 550–562, 2022.
- [20] L. Breiman, "Random forests," *Mach. Learn.*, vol. 45, no. 1, pp. 5–32, 2001.
- [21] C. Lei et al., "A comparison of random forest and support vector machine approaches to predict coal spontaneous combustion in gob," *Fuel*, vol. 239, pp. 297–311, 2019.
- [22] I. A. Ibrahim and T. Khatib, "A novel hybrid model for hourly global solar radiation prediction using random forests technique and firefly algorithm," *Energy Convers. Manage.*, vol. 138, pp. 413–425, 2017.
- [23] J. Bergstra and Y. Bengio, "Random search for hyper-parameter optimization," *J. Mach. Learn. Res.*, vol. 13, no. 2, pp. 281–305, Feb. 2012.
- [24] J. Mockus, *Bayesian Approach to Global Optimization: Theory and Applications*. Berlin, Germany: Springer, 2012, vol. 37.
- [25] Y. Lei, J. Lin, Z. He, and M. J. Zuo, "A review on empirical mode decomposition in fault diagnosis of rotating machinery," *Mech. Syst. Signal Process.*, vol. 35, no. 1/2, pp. 108–126, 2013.
- [26] Q. Miao, D. Wang, and M. Pecht, "Rolling element bearing fault feature extraction using EMD-based independent component analysis," in *Proc. IEEE Conf. Prognostics Health Manage.*, 2011, pp. 1–6.
- [27] M. H. Lipu et al., "Real-time state of charge estimation of lithium-ion batteries using optimized random forest regression algorithm," *IEEE Trans. Intell. Veh.*, 2022, doi: [10.1109/TIV.2022.3161301](https://doi.org/10.1109/TIV.2022.3161301).
- [28] J. Snoek, H. Larochelle, and R. P. Adams, "Practical Bayesian optimization of machine learning algorithms," in *Proc. 25th Int. Conf. Neural Inf. Process. Syst.*, Red Hook, NY, USA, 2012, vol. 2, pp. 2951–2959.
- [29] P. Nectoux et al., "Pronostia: An experimental platform for bearings accelerated degradation tests," in *Proc. IEEE Int. Conf. Prognostics Health Manage.*, 2012, pp. 1–8.
- [30] I. A. Ibrahim, M. Hossain, and B. C. Duck, "An optimized offline random forests-based model for ultra-short-term prediction of PV characteristics," *IEEE Trans. Ind. Informat.*, vol. 16, no. 1, pp. 202–214, Jan. 2020.
- [31] A. Liaw et al., "Classification and regression by RandomForest," *R News*, vol. 2, no. 3, pp. 18–22, 2002.
- [32] B. Tajiani and J. Vatn, "RUL prediction of bearings using empirical wavelet transform and Bayesian approach," in *Proc. 31st Eur. Saf. Rel. Conf.*, 2021, pp. 2006–2013.
- [33] A. Krogh, "What are artificial neural networks?," *Nature Biotechnol.*, vol. 26, no. 2, pp. 195–197, 2008.
- [34] M. Awad and R. Khanna, "Support vector regression," in *Efficient Learning Machines*. Berlin, Germany: Springer, 2015, pp. 67–80.
- [35] L. Meier, S. Van De Geer, and P. Bühlmann, "The group LASSO for logistic regression," *J. Roy. Stat. Soc.: Ser. B (Stat. Methodol.)*, vol. 70, no. 1, pp. 53–71, 2008.



Muhammad Gibran Alfarizi received the B.Sc. degree from the Bandung Institute of Technology, Bandung, Indonesia, in 2017, and the M.Sc. degree from the Norwegian University of Science and Technology, Trondheim, Norway, in 2020, both in petroleum engineering. He is currently working toward the Ph.D. degree in reliability, availability, maintainability, and safety with the Norwegian University of Science and Technology.

His research interests include safety and reliability, data-driven monitoring and optimization, and predictive maintenance.



Bahareh Tajiani received the bachelor's degree in industrial engineering from the University of Tehran, Tehran, Iran, in 2016, and the master's degree in reliability, availability, maintenance, and safety (RAMS) from the Norwegian University of Science and Technology (NTNU), Trondheim, Norway, in 2019. She is currently working toward the Ph.D. degree with RAMS group, NTNU.

Her main research interests include stochastic modeling, accelerated life tests, and signal

processing.



Jørn Vatn received the M.Sc. degree in mathematical statistics and the Ph.D. degree with a dissertation on maintenance optimization from the Norwegian University of Science and Technology (NTNU), Trondheim, Norway, in 1986 and 1996, respectively.

He has more than 30 years of experience as a Researcher with SINTEF. He is currently a Professor with Department of Mechanical and Industrial Engineering, Norwegian University of Science and Technology, Trondheim, Norway.



Shen Yin (Senior Member, IEEE) received the M.Sc. degree in control and information system and Ph.D. (Dr.-Ing.) degree in electrical engineering and information technology from the University of Duisburg-Essen, Duisburg, Germany, in 2007 and 2012, respectively.

He is currently a Full Professor (DNV-GL Professor) with the Department of Mechanical and Industrial Engineering, Norwegian University of Science and Technology, Trondheim, Norway. His research interests include fault diagnosis/prognosis and fault-tolerance strategy, safety, reliability, and security of complicated systems, system and control theory, data-driven and machine learning approaches, and applications in health diagnosis and cyber-physical systems.